Bank Credit Default Prediction:
A Comparative Study of Tabular ML Models and Metrics

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Problem statement

Credit default prediction is a **binary classification task** on tabular data with **high dimensionality, missing values and strong class imbalance**.

Model performance depends strongly on how carefully the pipeline handles missingness, encoding and data leakage.

Objectives

- Compare LogReg, SVM, Decision Tree models, AdaBoost, Random Forest, Gradient Boosting, XGBoost and MLP.
- Evaluate models using discrimination metrics: Precision, Recall, F1-score, ROC-AUC.
- Assess whether complex models will outperform interpretable baselines.

Existing solutions

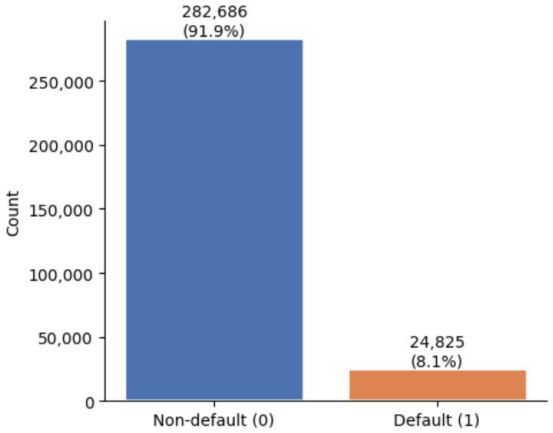
Logistic Regression:
industry standard, interpretable, underfits nonlinear interactions.

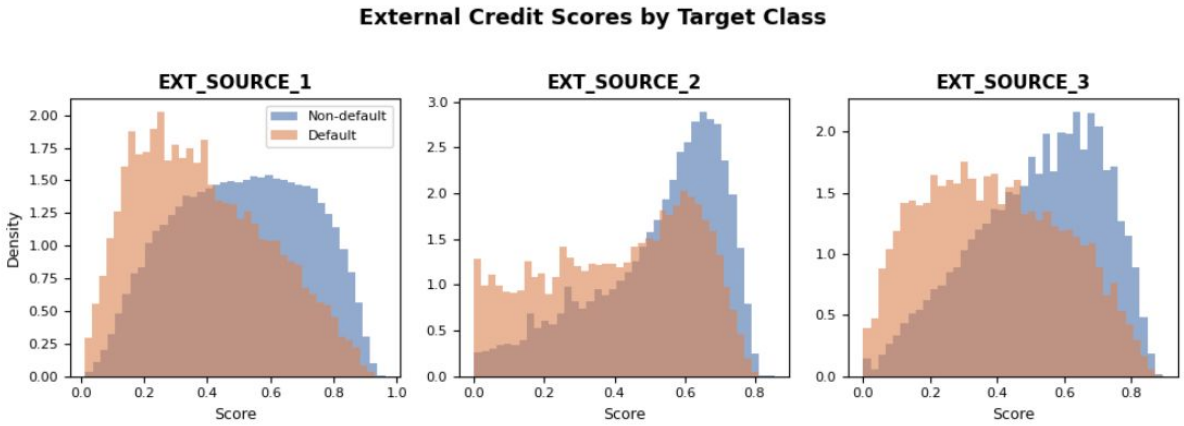
Ensemble/Boosting methods:
outperform single models in benchmark (Lessmann et al.).

Deep Learning:
sensitive to preprocessing and validation design,
limited adoption in regulated settings

S. Lessmann et al. “Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research,” *European Journal of Operational Research*, vol. 247, no. 1, pp. 124–136, Nov. 2015, doi: 10.1016/j.ejor.2015.05.030.

EDA revealed class imbalance in provided dataset





Dataset contains 307,511 loan applications, **only ~8% defaulted**.

⇒ Can't use accuracy for evaluation.

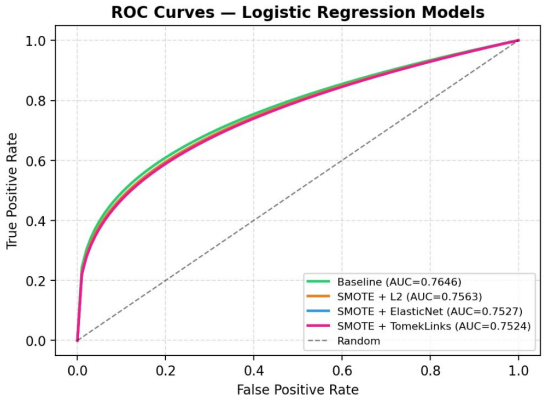
⇒ **Should use ROC-AUC, Precision and Recall**

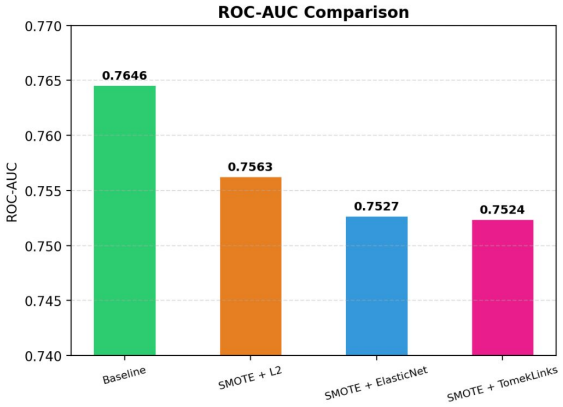
External Credit Scores are the strongest predictors.

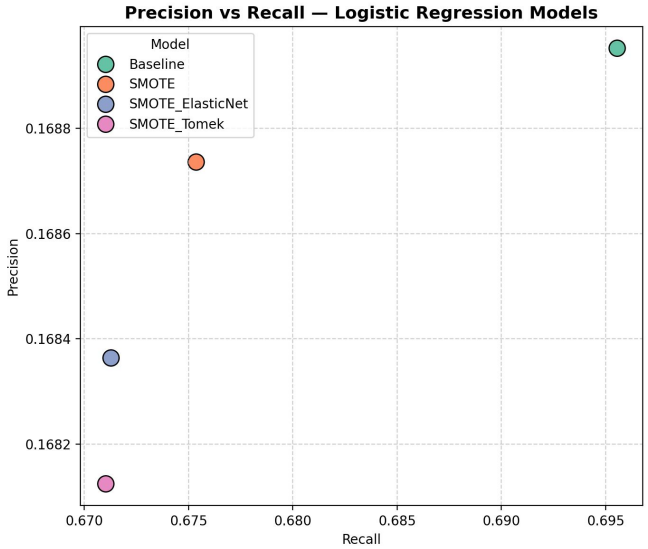
Clients with low scores are **more likely to default**, while high scores strongly reliable repayment.

Separation between classes visible across all three sources, so that these features and their aggregates are important features.

LogReg







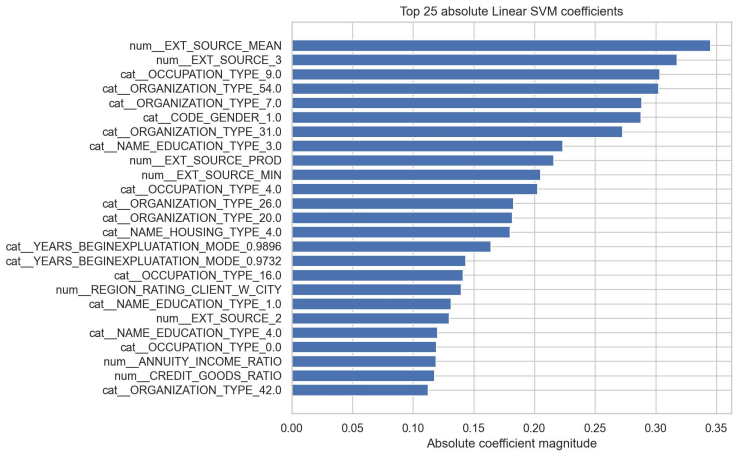
Baseline model (L2, C=0.1) achieves the best **ROC-AUC=0.7646,** outperforming all SMOTE-based models.

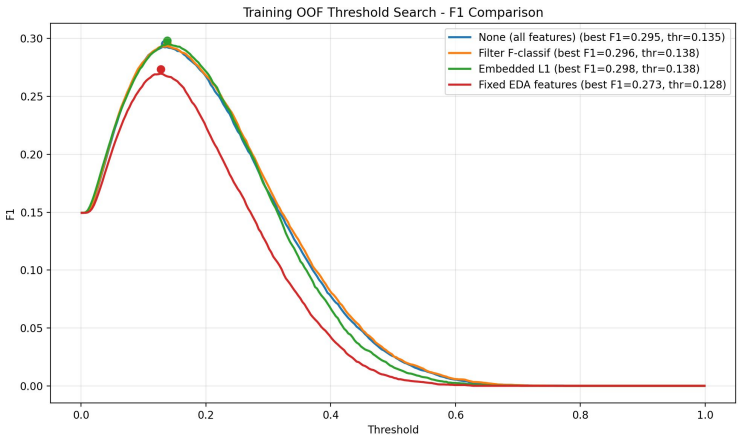
Resampling slightly improved Accuracy but reduced ROC-AUC and Recall. This confirms that **class_weight = 'balanced' is more effective and efficient** strategy for handling class imbalance in logistic regression.

Baseline model

Accuracy = 0.7646
Precision = 0.1690
Recall = 0.6955
F1 = 0.2719
ROC-AUC = 0.7646

SVM



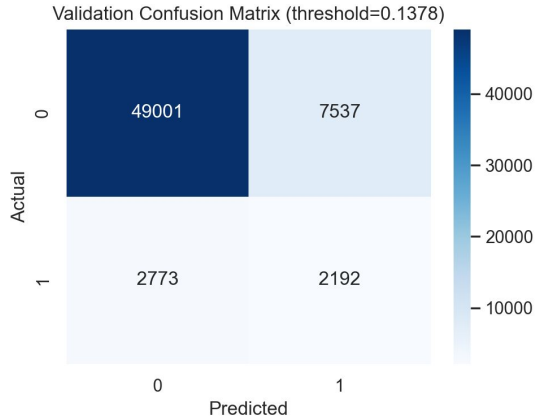


Best tuned SVM = Embedded L1
Accuracy = 0.832,
Precision = 0.225,
Recall = 0.441,
F1 = 0.298
ROC-AUC = 0.750,
Tuned threshold ≈ 0.138

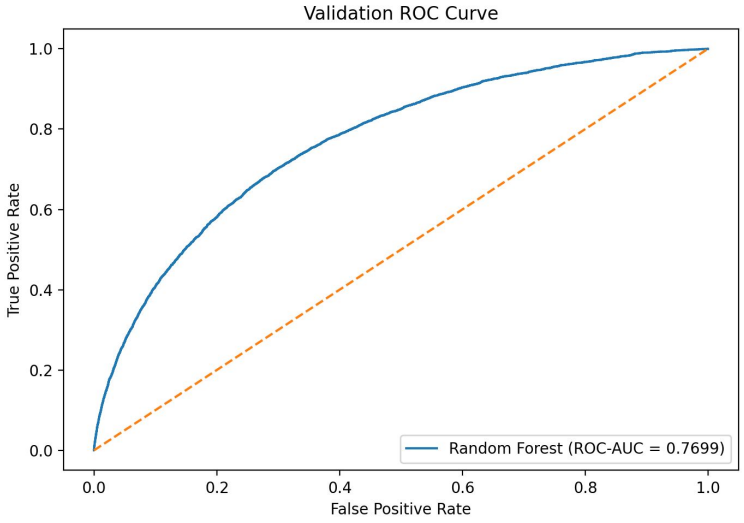
Compared four SVM settings:

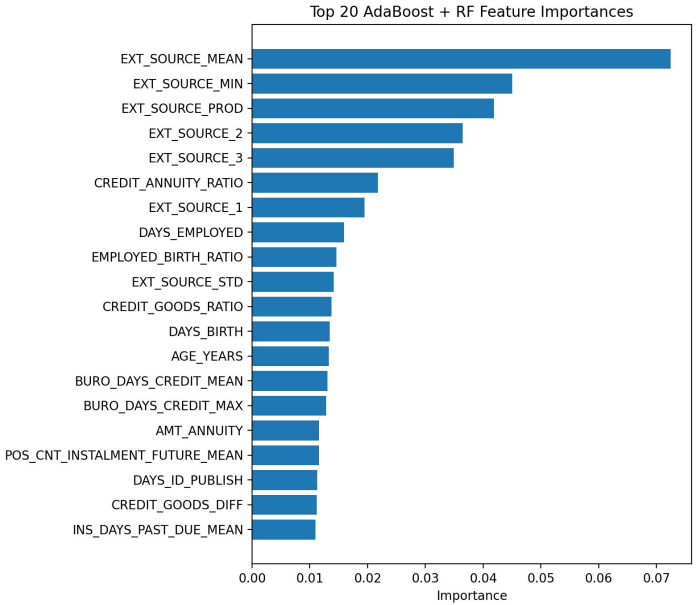
1. Full features,
2. Filter F-classif (ANOVA)
3. Embedded L1 selection
4. Fixed EDA feature subset.

Hyperparameters tuned with Optuna using 3-fold CV
Sigmoid calibration and OOF threshold search were applied to maximize F1 under class imbalance.



Decision Tree Models, Adaboost, And Random Forest





➢ **AdaBoost and Random Forest:** Balancing Precision and Recall

- AdaBoost with shallow decision trees as the base learners and Random Forest account for class imbalance using weights.
- External credit metrics, particularly EXT_SOURCE_MEAN, are the most important features, contributing to high F1 and ROC-AUC scores.

➢ **Combining Boosting and Random Forest:** Advantages and Limitations

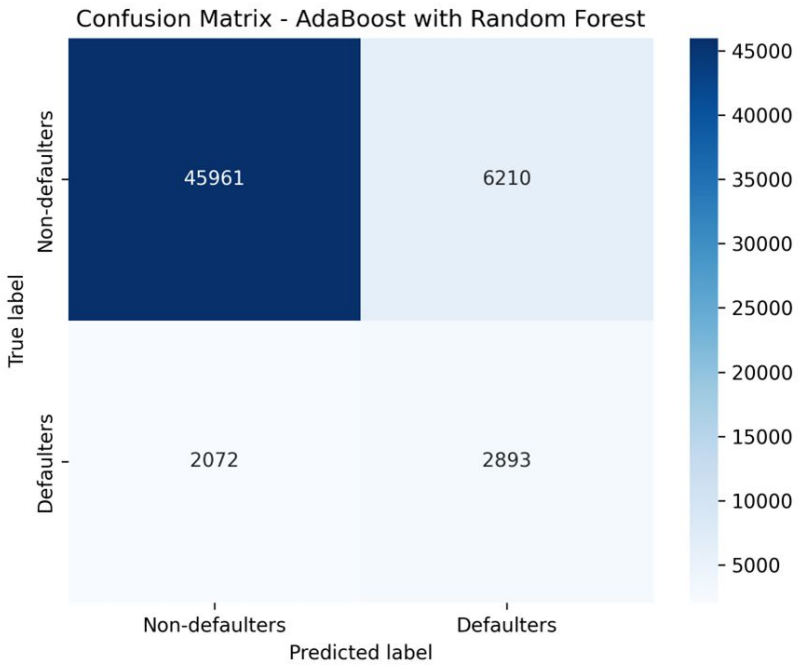
- AdaBoost + Random Forest model demonstrated the best F1 value (0.3326), with minimal gains compared to Random Forest.
- The computational cost and modest improvement require careful consideration of the model for practical use.

Comparison of the best models: ROC-AUC and F1

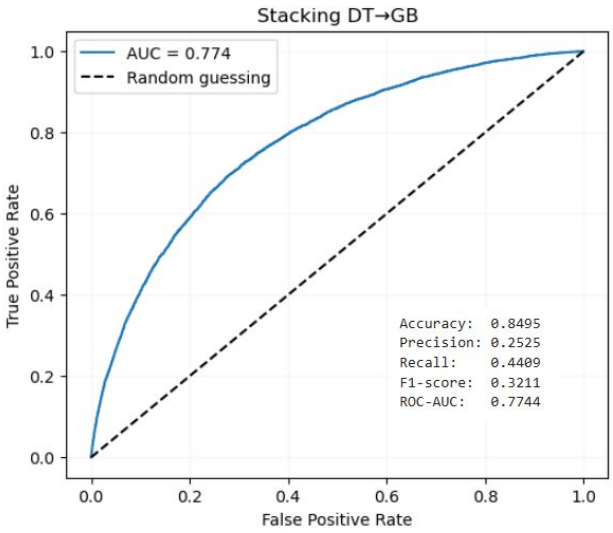
- Boosting ensembles and random forest show comparable **high ROC-AUC**, with boosting performing **better in F1**, which is important in the case of class imbalance

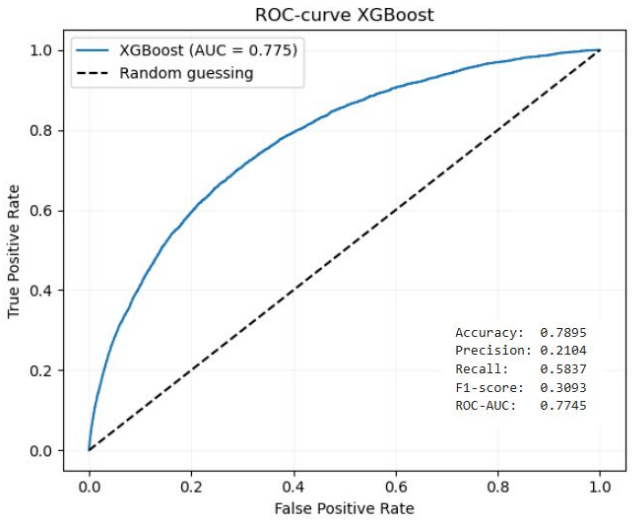
- **Best obtained metrics are for RF + Adaboost**

Accuracy = 0.8606;
Precision = 0.2711;
Recall = 0.4304**;**
F1-score = 0.3326**;**
ROC-AUC = 0.7726



Decision Tree with Gradient Boosting and XGBoost



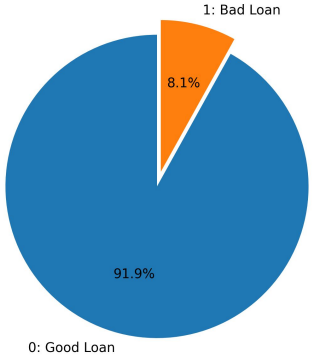


- Strong method applying 2 models to tackle with variance and bias
- Smote is used for the class balancing
- Max_depth is limited and unique threshold applied

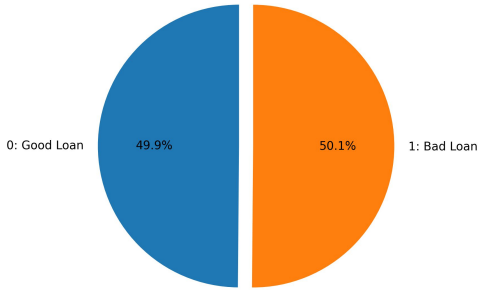
- One of the most powerful methods in practice
- Parameters are optimized by RandomizedSearch

Multilayer Perceptron (MLP) has shown the **best Recall** and **F1-score**

Initially



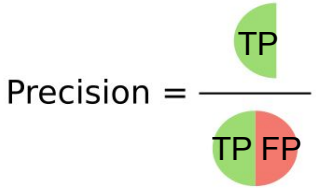
After SMOTE

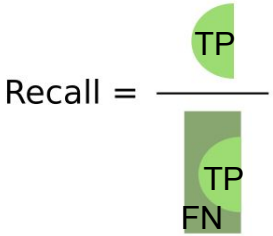


Best obtained metrics:

Accuracy =0.5245
Precision = 0.5145
ROC-AUC = 0.5262

Recall = 0.9161
F1-score = 0.6589





FN = 0 instead of 1 =>
bank gives money and lose it!

Recall is more important

Which model is the best?

Method	Accuracy	Precision	Recall	F1	ROC-AUC
LogReg	0.6992	0.1690	0.6955	0.2719	0.7646
SVM	0.8324	0.2253	0.4415	0.2984	0.7503
Decision Trees + AdaBoost	0.8687	0.2768	0.3881	0.3231	0.7719
RF	0.8538	0.2574	0.4298	0.3219	0.7699
RF with AdaBoost	0.8606	0.2711	0.4304	0.3326	**0.7726**
DT with Gradient Boost	0.8495	0.2525	0.4409	0.3211	**0.7744**
MLP	0.5245	0.5145	**0.9161**	**0.6589**	0.5262
XGBoost	0.7895	0.2104	0.5837	0.3093	**0.7745**

Conclusions

- The best **ROC-AUC** scores (0.77-0.78) were obtained on **Decision Trees** with different **boosting** techniques;

- **Recall** and **F1-score** are the highest in **MLP** results (0.91 and 0.67 respectively);

- The current work underlines, that even powerful modeling techniques cannot guarantee all metrics to be high while dealing with the problem of **imbalanced** data.

Thank you!

Any questions?

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